

Roll No. 47 / Experiment 10: Query Knowledge Graphs Using Cypher

Department of Computer Engineering - Knowledge Engineering Virtual Laboratory

Student Name:	Student Name	Student Roll / ID:	Roll No. 47
Experiment Date:	2026-09-18	Quiz Evaluation:	0 / 10 (0%)

1. Learning Objectives

- Retrieve entities (nodes) and their attributes from a knowledge graph using Cypher.
- Query direct semantic connections between entities via one-hop relationship traversal.
- Filter graph paths using boolean conditions on node properties with the WHERE clause.
- Traverse complex multi-hop paths to discover indirect relationships and convergent patterns.
- Perform aggregation using count() with implicit group-by semantics.
- Synthesize query observations, evaluate execution metrics, and document conclusions.

2. Knowledge Graph Domains Evaluated

1. University Graph (10 nodes: Students, Courses, Teachers, Departments; 8 relationships: STUDIES, TEACHES, BELONGS_TO)
2. Movie Graph (10 nodes: Actors, Directors, Movies, Genres; 9 relationships: ACTED_IN, DIRECTED, HAS_GENRE)
3. E-Commerce Graph (9 nodes: Customers, Products, Categories; 10 relationships: PURCHASED, BELONGS_TO)

3. Recorded Experimental Query Trials

No Cypher simulation trials recorded during this session.

4. Experimental Observations & Traversal Analysis

The experimental trials successfully verified that declarative Cypher pattern queries (MATCH, WHERE, RETURN) effectively retrieve nodes, relationships, and multi-hop paths from Knowledge Graphs without requiring relational JOINS. Traversal across directed edges and aggregations via count() demonstrated consistent execution behavior.

5. Conclusion

The experiment demonstrated how Cypher can be used to query knowledge graphs by retrieving nodes, relationships, filtered information, and multi-hop connections. The interactive traversal helped visualize how graph patterns are matched and how connected entities can be used to retrieve meaningful information.

Instructor / Student Signature